

of a bee, and can sense and respond dynamically to the environment. The colony's focus is on coordinating the behaviour of many independent robots so that these act as an effective unit.

Applications of RoboBees could include crop pollination, search and rescue missions, surveillance, as well as high-resolution weather, climate and environmental monitoring.

Invisible solar cells

See-through solar materials that can be applied to windows represent a massive source of untapped energy and could harvest as much power as bigger, bulkier rooftop solar units—Michigan State University (MSU) scientists report in *Nature Energy*. MSU scientists have developed a transparent luminescent solar concentrator that when placed on a window creates solar energy without disrupting the view. The thin, plastic-like material can be used on buildings, car windows, cell phones or other devices with a clear surface.

The solar-harvesting system uses organic molecules developed by the team to absorb invisible wavelengths of sunlight. The researchers can 'tune' these materials to pick up just the ultraviolet and near-infrared wavelengths, which then convert this energy into electricity.

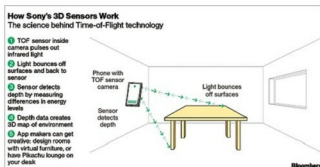


Highly transparent solar cells represent the wave of the future for new solar applications (Photo by Kurt Stepnitz/Michigan State University)

The authors note that there is an estimated 5 billion to 7 billion square metres of glass surface in the United States. And with that much glass to cover, transparent solar technologies have the potential to supply some 40 per cent of energy demand in the US—about the same potential as rooftop solar units. Currently, these exhibit efficiencies above 5 per cent, as against traditional solar panels' 15-18 per cent. Although transparent solar technologies will never be more efficient at converting solar energy to electricity than their opaque counterparts, these can get close and offer the potential to be applied to a lot more additional surface area.

3D sensors that can see the world

According to a Bloomberg report, Sony researchers are developing sensors that can detect people and objects by calculating how long it takes for light to reflect off surfaces. With plans to go into mass production next year, these sensors are designed for smartphones and augmented-reality devices. Eventually, these could find their way into drones, self-driving automobiles, gaming consoles, industrial equipment, factory and warehouse robots, and many other machines that interact with their environments.



How Sony's 3D sensors work (Source: www.bloomberg.com)

The new 3D detectors are in a category called time-of-flight (TOF) sensors, which scatter infrared light pulses to measure the time it takes for them to bounce back. The basic technology has been around for a while and forms the basis for the Xbox's motion-based Kinect, as well as laser-based rangefinders on autonomous vehicles and in military planes. Sony's big innovation over existing TOF sensors is that these are smaller and calculate depth at greater distances. Used with regular image sensors, these effectively give machines the ability to see like humans.

3D printer makes first wearable 'battery'

Imagine printing off a wristband that charges your smartphone or electric car with cheap components from a local hardware store. That's what Brunel University London scientists aim to achieve. They have already become the first to simply and affordably 3D print a flexible, wearable 'battery'. The technique opens the way for super-efficient, wearable power for phones, electric cars, medical implants like pacemakers and more.

The printer squirts stacks of silicone, glue and gel electrolyte pastes like a layered cake, to make what looks like a wristband. Sandwiched inside is a supercapacitor, which stores energy like a battery, but on its surface and without chemical reactions.